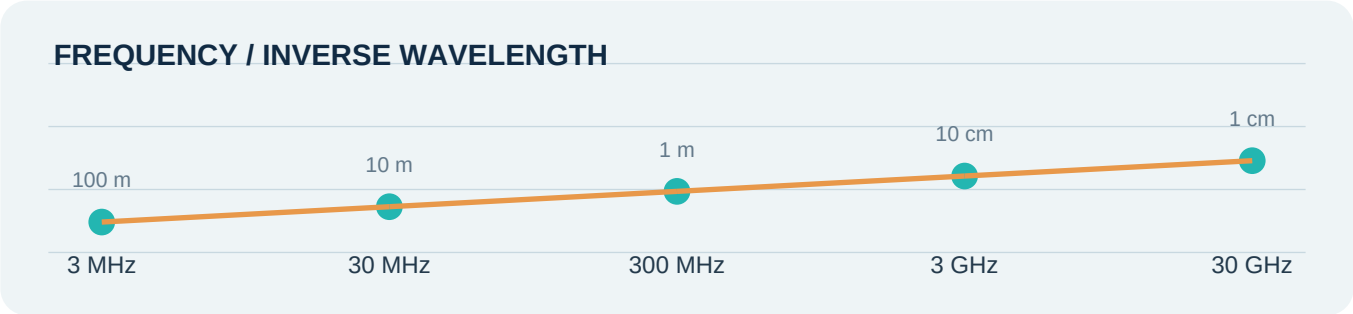
# Waves, frequency and wavelength

A physics-first illustrated guide

This original Atlas teaching brief introduces scientific foundations and links to primary literature. It is not a third-party research paper, peer-reviewed original result, or replacement for source documents.




## Why this matters

A radio signal is a measurement of an electromagnetic phenomenon. The physical relationship between frequency, wavelength, propagation and measurement uncertainty is the starting point for any scientific claim.

## Physical relation

In vacuum, lambda = c/f, where c = 299,792,458 m/s, f is cycles/second and lambda is distance/cycle. At 100 MHz, wavelength is about 3 m. In a material, the phase velocity may differ and depend on frequency.

## Measurement vs allocation

A frequency band is a way to classify a range. A regulatory allocation indicates permitted uses in a jurisdiction. Neither identifies an active source of radio energy nor proves a signal was observed.

## Model validity

Every propagation claim should specify environment, antenna model, geometry, polarization, uncertainty and calibration. Treat a convenient free-space approximation as an assumption, not a universal truth.

## Demonstration

Make a table of 3 MHz, 30 MHz, 300 MHz, 3 GHz, 30 GHz. Compute vacuum wavelengths and plot log frequency against log wavelength. The slope is minus one.

## Sources and further reading

See PUB-005: Electromagnetic Signal and Information Theory (arXiv:2605.12910), PUB-007: Smart Radio Environments (arXiv:2004.09352), and the Signal Atlas Science Division. Both arXiv PDFs are link-only under their submission licenses.